Deduce before you label: strategy attribution for large language model agents in the iterated prisoner's dilemma

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Abstract

Language models are often said to play tit-for-tat, always defect, or win-stay lose-shift, and such labels now feed models of mixed human-machine populations. A label of this kind claims something about the process behind a run of play, yet the usual procedures return one name for every input, whatever that input contains. We pass 4,800 agent-games from four models in the iterated prisoner's dilemma through a read-out with four answers. It asks by exact matching whether one canonical rule reproduces the whole run of play, and reports the set when several rules do. Where no rule fits it falls back on a classifier, and it returns no label at all when that classifier is unsure. On synthetic play the exact-matching stage is almost always right, and the abstention removes more than half of the error that is left. On our corpus a rule is proved for a quarter of the agent-games, nearly all of it unconditional cooperation or defection. A *reactive* rule is proved in 2.0% of agent-games but assigned fifteen times more often, and that gap survives controls for the length of the game. The mix is not a property of the agent either: multiplying every payoff by a positive constant cannot change the game, yet it moves the mix by up to 0.34 in total variation. Mining the unnamed quarter with a wider set of rules recovers two-phase play that opens with a probe, and behind it drift. We propose a reporting standard for strategy labels.

Keywords: machine behaviour; large language models; iterated prisoner's dilemma; strategy identification; selective classification; evolutionary game theory; multi-agent systems

1 Introduction

Large language models (LLMs) increasingly act as *agents* that negotiate, coordinate and decide on behalf of their users [1, 2]. Their behaviour in that role comes from strategic interaction rather than from single outputs. Training, prompting, role assignment and the language of the prompt all shape it, and each of them matters for safety and governance. The call to study machines as behaving entities, with the observational tools of ethology and the experimental discipline of the behavioural sciences, has become far more urgent as the population of interacting artificial agents has grown [3].

The iterated prisoner's dilemma is the natural test bed for that programme, because its strategy space is already mapped. A two-by-two matrix with $T > R > P > S$ and $2R > T + S$

defines reciprocity, forgiveness, willingness to punish and exploitation precisely, and evolutionary theory computes the fate of each of them [4–9]. The same matrix carries work on the statistical physics of cooperation [10–12] and on mechanisms such as intention recognition and commitment, which move the cooperative equilibrium [13, 14]. Because the map already exists, importing it is tempting, and a fast-growing literature now studies LLMs through game-theoretic lenses. That work reports departures from Nash equilibria, cooperative biases, model-specific signatures and archetypes recovered from long games [15–19]. Alongside it sit probes that use the instruments of cognitive psychology [20] and studies that treat models as simulated economic subjects [21–23].

The output of that literature is, very often, a name. Model X plays tit-for-tat; model Y is a defector; model Z looks like win-stay lose-shift. A name of this kind is not a summary statistic. It is a claim about the process that produced the play, and it allows conclusions that no rate allows. The agent will retaliate; it will forgive; we may place it in a replicator equation and compute its fate. The claim is therefore worth exactly as much as the procedure that produced it, and the procedures now in use share a property that is rarely stated. A classifier returns its highest-scoring label for every input, and a nearest-archetype distance returns a nearest archetype for every input. Neither can answer "none of these". Neither separates a run of play that a rule reproduces exactly from a run that merely lies closer to that rule than to the other three.

This paper takes strategy attribution itself as its object of study. We build a read-out that can return four different answers rather than one, apply it to a fully crossed corpus of 2,400 dyads, and ask three questions.

1. How much LLM play in the iterated prisoner's dilemma can be *shown*, rather than guessed, to follow a canonical strategy, and which strategies survive that test?

2. Is the resulting distribution over strategies a property of the model, or does it move with features of the prompt that leave the game unchanged?

3. What is the play that the read-out refuses to name, and does it hold a strategy that the canonical vocabulary is missing?

The questions are answerable because the first stage of the read-out is deductive. All four canonical rules are memory-one rules. We can therefore ask whether a run of play is exactly what AllC, AllD, tit-for-tat (TFT) or win-stay lose-shift (WSLS) would have played, and that question is a lookup rather than a guess. Only a unique match supports the statement that the agent played that rule. Several rules matching is an answer rather than a failure, because an agent that cooperates for ten rounds against a cooperator *is* AllC and TFT and WSLS at once. Only where that stage returns nothing does a long short-term memory (LSTM) classifier [24] supply a nearest rule, and only when its confidence clears a fixed floor. We report the four answers separately throughout and never pool them.

Our contributions are the following. **(i)** We build a strategy read-out that separates deduction from inference and is allowed to abstain. We validate it on synthetic play, where the generating rule is known, and on a held-out rule that shows what an unseen strategy does to a classifier that cannot abstain (§3.1). **(ii)** We report a census showing that deduction reaches a quarter of the corpus, that 92% of that quarter is constant play, and that reactive rules are labelled fifteen times more often than they are demonstrated (§3.2–§3.3). **(iii)** We show that the strategy mix moves beyond a permutation null under every change we apply, including a positive rescaling of the payoffs whose irrelevance is a theorem (§3.4). **(iv)** We mine the set the read-out will not name and recover a two-phase, switch-once structure that the canonical vocabulary has no room for, together with the evidence that the rest of that set is drift (§3.5–§3.7).

2 Methods

2.1 The game, the corpus and the changes we apply

FAIRGAME (Framework for AI Agents Bias Recognition using Game Theory) [25] provides the computational infrastructure. Conditions are written in JSON. At run time FAIRGAME combines them with language-specific prompt templates, simulates repeated normal-form games and logs every round of play. We adopt the framework and add payoff scaling. We take its multilingual templates as given rather than writing our own, which keeps the wording of the manipulations out of our hands.

Agents played a symmetric two-player prisoner's dilemma (figure 1a). The templates state the cell values as *penalties*, that is, years of imprisonment in the prisoner's-dilemma story, and they tell the agent to *minimise* them, so a smaller number is a better outcome. Under that framing the base cells are $T = 0$, $R = 2$, $P = 6$ and $S = 10$, which satisfies $T < R < P < S$. That ordering is the penalty-space image of the usual $T > R > P > S$. Option A is therefore the dominant, defecting action and Option B is cooperation. On the utility scale $u = -\text{penalty}$ the two normalised measures of the dilemma are greed $(T - R)/(T - S) = 0.20$ and fear $(P - S)/(T - S) = 0.40$.

We state the direction of the payoffs explicitly, because turning it around would swap the AllC and AllD columns of every table below. The assignment is not a matter of interpretation. FAIRGAME logs the realised payoff next to every action. The assignment above reproduces the logged payoff on all 24,000 rounds of the corpus, and the opposite assignment reproduces none of them.

We crossed three factors on top of that fixed game (figure 1b). First, we multiplied every payoff cell by a scale $\lambda \in \{0.1, 1, 10\}$. Because $\lambda > 0$, the operation is a positive affine rescaling of a von Neumann–Morgenstern utility [26, 27], so the ordering of outcomes, the best-reply correspondence, the equilibrium set and the replicator dynamics all stay the same. The three games are one game shown with different numbers. Second, we ran the same game in English, French, Vietnamese, Chinese and Arabic, since a game does not change with the language its rules are written in. Third, each agent received a system-prompt persona, either *cooperative* or *selfish*, and we crossed the four ordered pairings with everything else. A persona is not irrelevant by theorem, but it carries no information about the game: the four pairings present identical payoff matrices, and an agent was never told its opponent's persona. We recovered the realised scale of each game from the logged values rather than assuming it from the directory name.

The design crossed four models, Claude 3.5 Haiku, Gemini 3.5 Flash-Lite, GPT-4o and Mistral Large. Each model met three payoff scales, five prompt languages, four persona pairings and ten replicates, which gives $4 \times 3 \times 5 \times 4 \times 10 = 2{,}400$ dyads, 600 per model. Each dyad ran for ten rounds and we record both agents, which gives 4,800 agent-games and 48,000 binary decisions, with exactly 40 dyads in every model $\times$ scale $\times$ language cell. Every decision resolves to one of the two option tokens and every dyad ran its full ten rounds, so no cell loses games to a failure to parse. One GPT-4o dyad is logged with a mixed persona pairing where the design specifies a matched one, so that model's persona margins are 599 and 601. No conclusion below turns on that single dyad.

On each round an agent received the full payoff table, its persona, the round index and the complete history of both players' choices, and it returned a single token naming one of two options. We disabled communication, so the only channel between agents is the action history. Actions carry the labels `OptionA` and `OptionB` in all languages, which avoids the extra meaning that "cooperate" and "defect" would bring. The templates never say which label means cooperation; the agent has to read it off the payoff table.

Decoding used a fixed low output-token budget with reasoning traces disabled. That setting is not a detail. A model that emits hidden reasoning can use up a small budget inside the trace and return an incomplete string. A harness that then falls back to a default action turns that string into an apparent cooperation rate of 1.0, which the parser has created rather than the model. We logged parse-failure and fallback rates for every cell and required both to stay below 0.02. Sampling temperature follows each provider's recommended default, as in the FAIRGAME protocol: 1.0 for Claude, Gemini and GPT-4o, and 0.3 for Mistral Large.

Two features of the design are confounded with model identity, and we declare them here rather than leave a reader to discover them. We ran the Gemini arm with the round count disclosed and the other three with the horizon hidden, so the Gemini column is a "model × horizon" column and figure 1b marks it as such. Sampling temperature, fixed by the provider default above, likewise cannot be separated from model identity. Both features are constant within a model, so neither can generate any within-model effect reported below, and only between-model comparisons carry them. The consequence, which we invoke wherever it bites, is that no read-out statistic is comparable *across* models when decoding stochasticity would itself move the comparison. The coverage figures of §3.2 and §3.5 are of that kind, so the claims we draw there are about the panel as a whole, not about its ordering.

2.2 A read-out with four answers

We encode a player's run of play as a sequence of two-letter tokens, ⟨previous-round outcome⟩⟨action this round⟩. The outcome letters are E (no history), R (both cooperated), S (the player cooperated and was exploited), T (the player exploited) and P (mutual defection). The synthetic corpus uses exactly the same encoding, which is what lets a classifier trained on generated play read a transcript of real play.

Stage one: deduction. Each canonical rule fixes an action for every outcome letter. AllC cooperates everywhere and AllD defects everywhere; TFT plays C after E, R, T and D after S, P; WSLS plays C after E, R, P and D after S, T. Asking whether a run of play is exactly what rule k would have played is therefore a per-token lookup, and it needs no learning at all. The matcher returns a *set*. Exactly one member supports the statement that the agent played that rule, and several members is the correct answer when the realised history cannot separate them.

Stage two: inference, with the option of an abstention. Where no rule fits, an LSTM classifier trained on synthetic play from the four known rules returns a posterior, that is, a confidence score, over the four rules. We take its highest-scoring label as the *nearest* rule, never as an identification. The network is a single-layer LSTM with a 24-dimensional embedding and 96 hidden units, trained on the union of the noise-free and 5%-execution-noise splits of the ten-round synthetic corpus. We retrain nothing on LLM play. We accept its label only when the top score reaches a floor of 0.90, and otherwise the read-out abstains.

The floor is a convention here rather than a discovery. On thirty-round play the confidence distribution has two clear modes and 0.90 falls in the gap between them. At the ten-round horizon of this corpus it does not, so we cannot defend the value as a valley. We defend it instead with the risk-coverage curve of §3.1, which we report in full, and we give the census at floors of 0.80, 0.90 and 0.95 so that no conclusion rests on where the cut falls.

Four answers therefore leave the read-out for every agent-game: *deduced* (exactly one rule reproduces the run of play), *rule set* (several rules do), *nearest rule* (none does and the classifier is confident) and *abstained* (none does and the classifier is not). We never pool the four (figure 1c). We also record the smallest number of rounds on which play departs from its own nearest rule, which measures how much work a nearest-neighbour label is being asked to do.

Grim trigger is a fifth memory-one rule, and we treat it asymmetrically. We flag the asymmetry because it runs against us: grim trigger enters every distance but not the exact-match vocabulary, so a run of play that only grim trigger would reproduce counts as unmatched. Admitting it raises the share of games matched by at least one rule by 0.0, 0.2, 3.3 and 0.6 percentage points for Claude, Gemini, GPT-4o and Mistral, and it lowers Gemini's uniquely matched share by 0.5 points. Grim trigger is memory-one, $(1, 0, 0, 0)$ opening with C, so it sits inside the 32-rule class of §3.5 throughout.

2.3 Validating the read-out where the truth is known

We generate the synthetic corpus by replaying each of the four rules against sampled opponents, with execution noise at 0%, 5%, 10% and 20%; the network sees only the first two levels. Because we record the generating rule, three quantities that no one can measure on LLM play are available there. The first is whether the deductive stage recovers the truth when it fires. The second is how accuracy grows with the number of rounds observed. The third is what the abstention buys. We report the risk-coverage curve, the accuracy of the deductive stage given that it fires, and the identifiability curve against its Bayes ceiling. The ceiling is the accuracy of an oracle that knows the generating distribution and is limited only by runs of play that several rules produce alike.

A fourth quantity needs a rule the network has never seen. Generous tit-for-tat, which forgives a defection with fixed probability, was held out of training entirely and serves as a positive control for novelty. Whatever the classifier does with generous tit-for-tat is what it will do with any strategy outside its vocabulary.

2.4 Widening the vocabulary, and the nulls

A rule may fail to fit because the vocabulary is too narrow rather than because the play holds no rule at all. We therefore mine the abstention set with three progressively wider hypothesis classes, each paired with a null. The first class is all 32 deterministic memory-one rules, that is, an opening move crossed with an action for each of the four conditioning states. The second admits runs of play built from two such rules with a single switch. The third is a library of fifteen named non-canonical strategies from the tournament literature, including suspicious TFT, grim trigger, tit-for-two-tats, gradual, prober, contrite TFT, soft and hard majority and the alternator. To that library we add the two-phase clock rules built by joining an ordered pair of opponent-only strategies. Only AllC, AllD and TFT can be joined in that way, because WSLS reads the player's own last action, which gives six ordered pairs at nine switch points and 69 candidate rules in all. That count is why the null matters: with 69 chances to fit a ten-round binary sequence, a coverage figure means nothing until the same search has been run on shuffled play.

We pair every coverage figure with a permutation null in which we shuffle the focal player's own action sequence in place, holding its cooperation rate and the opponent's realised sequence fixed. The null is not a formality. A two-segment fit over ten rounds has enough freedom to reproduce a large share of shuffled play by chance. Constant play, in addition, does not change under the shuffle at all. Raw coverage on its own is therefore not evidence of structure.

Two further probes ask what the play is without assuming any vocabulary. The first counts every four-round window of the focal player's own actions and compares its frequency with the same within-game shuffle, so a lift above one is temporal structure and nothing else. Alternation is the pattern a memory-one rule cannot express, so the CDCD and DCDC windows are the ones to read first. The second probe scores nested predictors of the next action by the Bayesian information criterion [28] on identical rows. The predictors are memoryless, memory-one, memory-two, and a memory-one rule allowed to change with the phase of the game; because the rows are identical, the comparison is about history rather than about sample size.

Finally, we ask whether the unnameable play is a strategy at all. Play generated by a fixed rule is coherent across a game by construction. We therefore split each game at round five and correlate the cooperation rate of the two halves within each answer class. We also cluster the abstention set in memory-one profile space, sweeping k from 2 to 8 and reporting the silhouette alongside a bootstrap adjusted Rand index.

2.5 Statistical analysis

All intervals are 95% bootstrap confidence intervals over 2,000 resamples that resample whole dyads [29]. The choice of unit matters. Rounds within a game are autocorrelated and the two agent-rows of a dyad share a history, so a row-level resample would make the interval too narrow by roughly a factor of $\sqrt{2}$.

We answer the question of whether a manipulation moves the strategy mix with a distributional test rather than with a contrast on a single share. For each model and factor we take the largest total-variation distance between the five-category mixes of any two levels. We compare that distance with a permutation null that shuffles the factor label across *dyads*, so a dyad keeps both of its agent-rows together, and we draw 2,000 permutations. We report the observed distance, the permutation P value and the null's 95th percentile, which doubles as the smallest shift the test could have detected.

We do not correct for multiple comparisons, and we say so rather than leave the point to be inferred. Each model is a separate system and is its own family, so the tests within a manipulation are four families of one rather than one family of four. The claims we draw are in any case about whether manipulations move the mix at all, not about which model moves most.

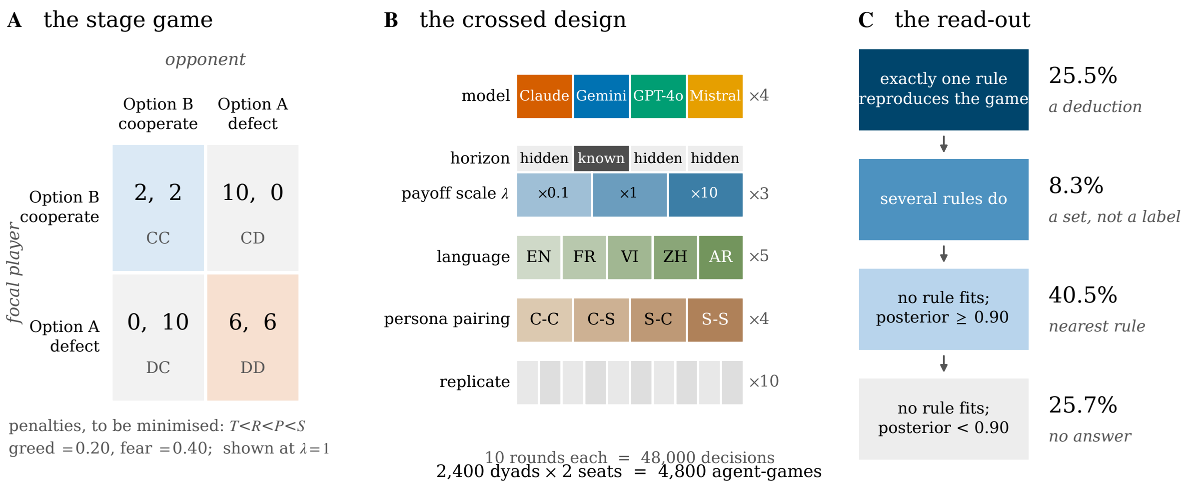

Figure 1: The game, the design and the read-out. (**a**) The stage game at $\lambda = 1$. Cells are penalties that the agent is told to minimise, so $T < R < P < S$ and Option A is the dominant, defecting action. Greed and fear are computed on the utility scale $u = -$penalty. Every cell is multiplied by $\lambda \in \{0.1, 1, 10\}$, a positive affine rescaling that leaves the game strategically identical. (**b**) The fully crossed design: 2,400 dyads, 4,800 agent-games, 48,000 binary decisions, with 40 dyads in every model × scale × language cell. The horizon row is shaded as a confound rather than a factor, because only the Gemini arm was told the round count. (**c**) The read-out and what it returns on this corpus. The first two steps are deductive lookups over the four canonical memory-one rules and involve no learning. The third step is an LSTM trained on synthetic play, admitted only above a confidence floor, and the fourth step is an abstention. Shares are of all 4,800 agent-games.

3 Results

3.1 What the read-out is worth

The deductive stage of §2.2 is exact by construction, but exactness is not usefulness. The stage could fire so rarely that it is useless, or so often that it says nothing. On the synthetic test set of §2.3 it fires with a unique answer on 80.3% of runs of play and recovers the generating rule in 99.8% of them. A further 0.7% receive a set, which holds the truth 99.4% of the time, and the remaining 19.0% match nothing, which is what 5% execution noise does to an exact matcher over ten rounds. Deduction is therefore both frequent and near-certain when the play really is rule-generated. Those numbers are the benchmark against which the LLM figures of §3.2 should be read.

The deductive stage is also not biased towards the constant rules at this horizon, which matters for what follows. Resolved by generating rule, it returns a unique answer on 79.7% of AllC-generated runs, 80.9% of AllD, 80.8% of TFT and 79.6% of WSLS, and where it fires it is right 99.8%, 100.0%, 99.6% and 99.7% of the time. Ten rounds are enough to prove a reactive rule as often as a constant one, which rules out the reading that §3.3 is a statement about the horizon rather than about the play.

The classifier that covers the remainder needs ten rounds and no more (figure 2a). Accuracy rises from 0.487 after one round, where the opening move alone separates AllD from the three rules that open cooperatively, to 0.984 after ten, against a Bayes ceiling of 0.993. The gap to the ceiling is 0.009. Whatever the read-out fails to do on LLM play is therefore not a failure of horizon or of capacity on this vocabulary. Under execution noise the classifier degrades gently

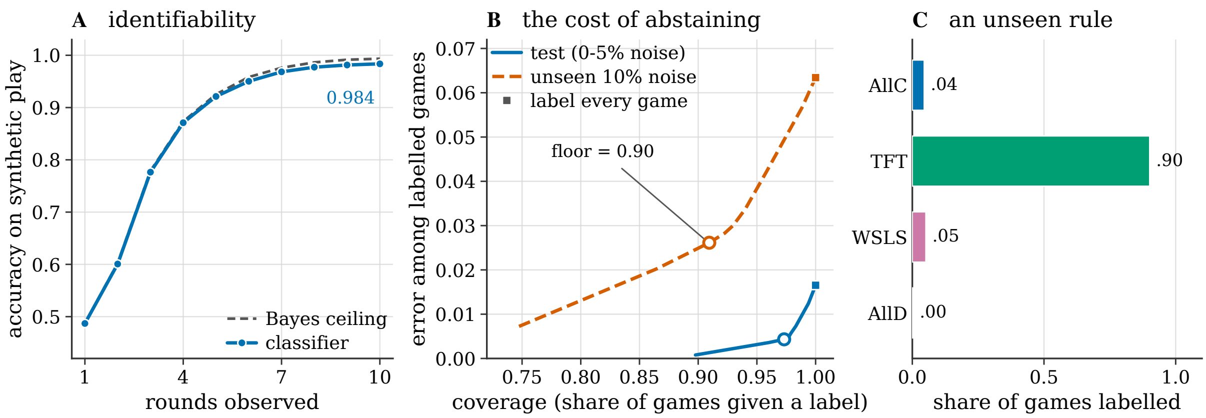

Figure 2: Validation on synthetic play, where the generating rule is known. (**a**) Accuracy as a function of the number of rounds observed, against the Bayes ceiling of an oracle that knows the generating distribution and is limited only by runs of play that several rules produce alike. Ten rounds are enough for the classifier to reach 0.984 against a ceiling of 0.993, so the horizon is not what fails on LLM play later. (**b**) The risk-coverage curve. Sweeping the confidence floor trades coverage for error among the games that still carry a label; squares mark labelling every game and rings mark the 0.90 floor used throughout. On play 10% of whose actions are noise, a rate the network never saw in training, abstaining on 9% of games removes 59% of the error. (**c**) What a classifier does with a rule it has never seen. Generous tit-for-tat was held out of training entirely; 90% of its games are labelled TFT and none is flagged as new. Any read-out that cannot abstain will absorb an unknown strategy into its vocabulary instead of reporting one.

inside its training range, from 0.995 at 0% to 0.973 at 5%, and less gently outside it, to 0.937 at 10% and 0.810 at 20%.

That degradation is what the abstention is for (figure 2b). At the 0.90 floor the read-out declines to label 2.7% of the low-noise test set, and error among the games it still labels falls from 1.65% to 0.43%. On the unseen 10%-noise split it declines 9.1% of games and error falls from 6.34% to 2.61%, a reduction of 59%. The floor is a chosen operating point on a curve rather than a threshold with independent standing, which is why we report the whole curve and repeat the census below at 0.80 and 0.95.

The held-out rule makes the sharpest case for having an abstention at all (figure 2c). Generous tit-for-tat never reached the network, and the network absorbs it: 90.1% of its games come back labelled TFT, 5.1% WSLS, 4.5% AllC and 0.3% AllD. Nothing in the output says "this is not one of my four". A read-out that must return a label will therefore report a vocabulary it was given rather than a strategy the agent played, and it will do so with no outward sign that anything is wrong. The behaviour is not a defect of this particular network; it is what forcing a single best guess means.

3.2 A census of named play

Applied to the corpus of §2.1, the read-out deduces a unique rule in 25.5% of agent-games, returns a set in 8.3%, supplies a nearest rule in 40.5% and abstains in 25.7% (figure 3a). The spread across models is large. Mistral Large is deduced in 48.7% of its games and Claude 3.5 Haiku in 9.7%, with Gemini 3.5 Flash-Lite and GPT-4o between them at 21.9% and 21.8%. We draw no conclusion from that ordering, because sampling temperature is confounded with model identity (§2.1). The panel-level statement is what matters, and it does not depend on the

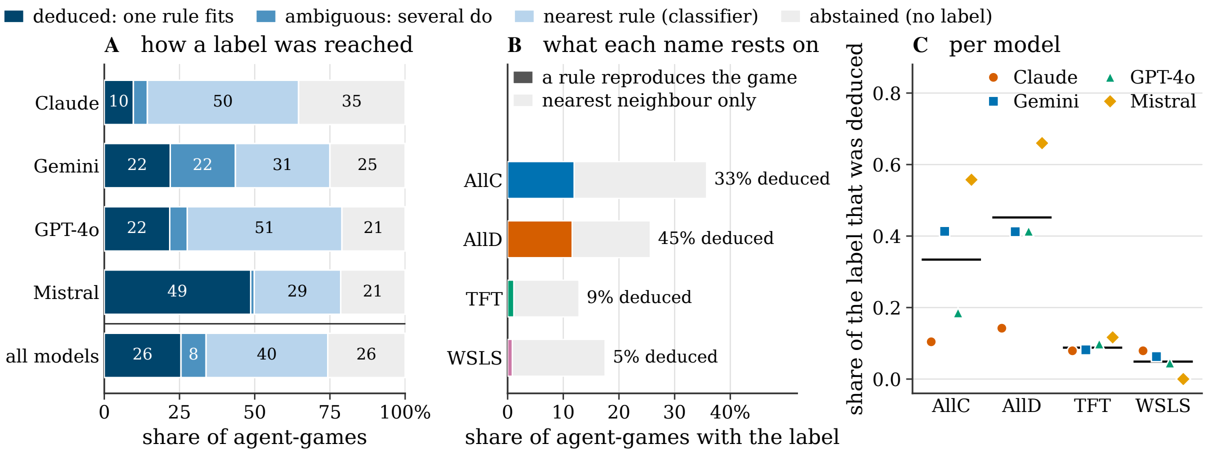

Figure 3: What a strategy label is made of. (**a**) The four answers per model and pooled. Dark: exactly one canonical rule reproduces the whole run of play. Mid: several rules do, and the ten rounds do not separate them. Pale: none does, and the classifier supplies a nearest rule. Lightest: none does and the classifier abstains. (**b**) The same corpus resolved by label rather than by game. Each bar is the share of agent-games carrying that name, split into the part a rule reproduces exactly and the part that is only a nearest-neighbour statement. AllC and AllD are largely deduced; TFT and WSLS are almost entirely inferred. (**c**) The same quantity per model. Bars mark the pooled value. Mistral Large receives the WSLS label in 14.3% of its agent-games and plays an exact WSLS game in none of them.

ordering. In every model, at least half of the play matches no canonical rule at all, from 50% for Mistral Large to 86% for Claude 3.5 Haiku.

The census is stable against the confidence floor in the direction that matters. Moving the floor from 0.80 to 0.90 to 0.95 moves the abstained share from 20.0% to 25.7% to 32.4% and moves nothing else, because the deductive stage runs first and is unaffected. The deduced and rule-set shares are 25.5% and 8.3% at every floor, and no claim below depends on where inside that band the cut falls.

The composition of the deduced quarter is the first main result. Of the 1,224 agent-games in which exactly one canonical rule reproduces the whole run of play, 1,129, or 92.2%, are AllC or AllD. A *reactive* rule, TFT or WSLS, is deductively established in 95 agent-games, or 2.0% of the corpus. Per model, the constant share of the deduced set is 68.1% for Claude, 90.1% for Gemini, 90.4% for GPT-4o and 98.8% for Mistral. What the deductive stage mostly proves, in other words, is that an agent did the same thing ten times. The rules that carry the theoretical content of the iterated prisoner's dilemma, the ones that condition on what the opponent did, are almost never proved.

One reading of that gap has to be excluded before anyone can believe it. A reactive rule differs from a constant one only when the opponent gives it something to react to: against an opponent that never defects, TFT and AllC prescribe the same ten actions and no method can separate them. We therefore restrict the corpus to the agent-games whose opponent history makes the difference visible, that is, games in which the opponent plays at least one C and at least one D in rounds one to nine. Those games are 66.1% of the corpus, ranging from 50.1% for Mistral Large to 88.4% for Claude. Inside them the deduced-reactive share rises only from 2.0% to 2.5%, while the deduced-constant share is 18.0%. The scarcity of proved reactive play is therefore not created by opponents who never gave these agents an occasion to reciprocate. It survives conditioning on exactly those occasions, at a horizon where the same matcher proves

TFT on four fifths of TFT-generated play (§3.1).

That composition also explains a result that would otherwise look bad for the deductive stage. Against the shuffled null of §2.4, canonical coverage is barely enriched: the excess is 1.8 percentage points for Claude, 1.5 for Gemini, 2.0 for GPT-4o and 0.4 for Mistral. The result could not be otherwise, because a constant run of play does not change when we permute its own actions. Every AllC and AllD match therefore survives the shuffle by construction, and those matches are 92% of all matches. The enrichment test is thus uninformative about constant play and informative about reactive play, where it says that reactive matches are rare in the data and in the null alike.

3.3 The labels that do not survive the audit

Resolving the same corpus by label rather than by game is where the two stages come apart (figure 3b). AllC is assigned to 35.7% of agent-games, and 33.4% of those assignments are deductions; AllD is assigned to 25.6% with 45.2% deduced. TFT is assigned to 12.8% of agent-games with 8.8% deduced, and WSLS to 17.5% with 4.9% deduced. Put the other way, 91.2% of TFT labels and 95.1% of WSLS labels in this corpus are nearest-neighbour statements about play that no rule in the vocabulary reproduces.

The gap is not created by a small denominator. Reactive labels go to 30.3% of agent-games and reactive rules are demonstrated in 2.0%, a factor of 15.3. Per model the ratio is 12.6 for Claude, 14.1 for Gemini, 15.6 for GPT-4o and 33.1 for Mistral Large. The WSLS label appears in 14.3% of the agent-games of Mistral Large and is deduced in none of its 1,200 (figure 3c). A statement that Mistral Large plays win-stay lose-shift is, on this corpus, a statement that its play was closer to WSLS than to the other three, in games none of which WSLS would have produced.

We can also measure what that closeness is worth. Where a label comes from the nearest-neighbour stage, play departs from the rule it names on 1.97 rounds out of ten for AllC, 2.01 for AllD, 2.49 for TFT and 2.09 for WSLS. A rule that is wrong on a fifth to a quarter of the decisions it is supposed to generate is not a description of the process. It is the closest of four available descriptions, three of which are also wrong.

Table 1 collects the per-model read-out together with the two quantities the rest of the paper turns on: the shift in the mix under payoff rescaling, and the coverage that the wider vocabulary buys on the abstained games.

The claim here is about evidence, not about the network. The classifier is accurate wherever accuracy is definable (§3.1), and its nearest-neighbour statements are not errors, because there is no truth for it to be wrong about. The error lies in reading such a statement as an identification. A single pooled label distribution invites that error, and reporting the four answers separately prevents it.

3.4 The strategy mix is a property of the prompt

A strategy label is meant to name something about the agent. If the distribution over labels is instead a function of how the game was written down, then a census measures the prompt.

It does (figure 4). Every one of the sixteen model × factor comparisons moves the strategy mix beyond the permutation null of §2.5. The sharpest case is the payoff scale, because there the null is not a convenient default but a theorem. Multiplying every cell by $\lambda > 0$ leaves best

Table 1: The read-out per model. Every model contributes 600 dyads, 1,200 agent-games and 12,000 decisions. "Deduced" is the share of agent-games in which exactly one of the four canonical rules reproduces the whole run of play; "constant" is the share of that deduced set which is AllC or AllD. "Reactive" compares the share of agent-games in which TFT or WSLS is deductively established with the share carrying one of those two labels by any route. "λ shift" is the largest total-variation distance between the strategy mixes at two payoff scales, whose null is exactly zero because the rescaling is a positive affine transformation; we give the null's 95th percentile for scale. The last column mines the abstained games with the extended vocabulary of §2.4 against its shuffled null. Coverage statistics are not comparable across models, because sampling temperature is confounded with model identity (§2.1).

	Coop.			Reactive rule			Abstained set named	
Model	rate	Deduced	constant	proved	labelled	Abstained	obs.	null
Claude 3.5 Haiku	0.63	9.7%	68.1%	3.1%	39.0%	35.4%	0.27	0.10
Gemini 3.5 Flash-Lite	0.74	21.9%	90.1%	2.2%	30.5%	25.0%	0.56	0.14
GPT-4o	0.45	21.8%	90.4%	2.1%	32.4%	21.0%	0.39	0.07
Mistral Large	0.52	48.7%	98.8%	0.6%	19.3%	21.4%	0.59	0.10
All models	0.58	25.5%	92.2%	2.0%	30.3%	25.7%	0.43	0.10

Every λ shift exceeds its permutation null: 0.263, 0.338, 0.325 and 0.125 against null 95th percentiles of 0.113, 0.123, 0.105 and 0.098 ($P < 0.001$ for the first three, $P = 0.006$ for Mistral Large).

replies, equilibria and replicator dynamics unchanged, so the mix of a strategic agent must not move at all. It moves by 0.263 for Claude, 0.338 for Gemini, 0.325 for GPT-4o and 0.125 for Mistral in total variation. The matching null 95th percentiles are 0.113, 0.123, 0.105 and 0.098, with $P < 0.001$ for the first three models and $P = 0.006$ for the fourth. Resolved into shares, GPT-4o carries the AllD label in 56.8% of its games at $\lambda = 0.1$, in 27.2% at $\lambda = 1$ and in 24.2% at $\lambda = 10$. The AllD share of Claude falls from 35.0% to 10.2% and that of Gemini from 23.2% to 2.0% across the same rescaling. Writing the same payoffs as 0.2 rather than 2 roughly halves the share of play that is recorded as unconditional defection.

Because "Ambiguous" is a statement about identifiability rather than about behaviour, a shift in the five-category mix could in principle be games moving into and out of that bin rather than a change in what the agents do. It is not. Repeating every test over the four named strategies alone leaves all sixteen comparisons above their nulls, with the payoff-scale distances at 0.245, 0.304, 0.296 and 0.127 and no P value above 0.0065.

Prompt language moves the mix in all four models as well, by 0.188 to 0.296 against nulls of 0.142 to 0.175. Here the null is a weaker claim than for the payoff scale, and one limitation of the test is a finding rather than a problem. The FAIRGAME templates do not state the objective identically in every language: English, French and Vietnamese tell the agent to minimise a penalty, while Chinese and Arabic tell it to maximise a reward. A five-language comparison therefore mixes a language manipulation with an objective-framing manipulation. We report the five-language test as the omnibus test it is, that is, as one test over all five languages at once, and we do not break it apart. The strategy mix is the dependent variable here, and the framing subset leaves too few dyads per cell to estimate a five-category distribution stably.

The weakest of the sixteen comparisons is the language test for Gemini, at 0.188 against a null 95th percentile of 0.175 ($P = 0.028$), and it is the one entry that a family-wise correction across the four models would absorb. It is also the one entry that the four-category re-test strengthens rather than weakens, to 0.205 at $P = 0.003$.

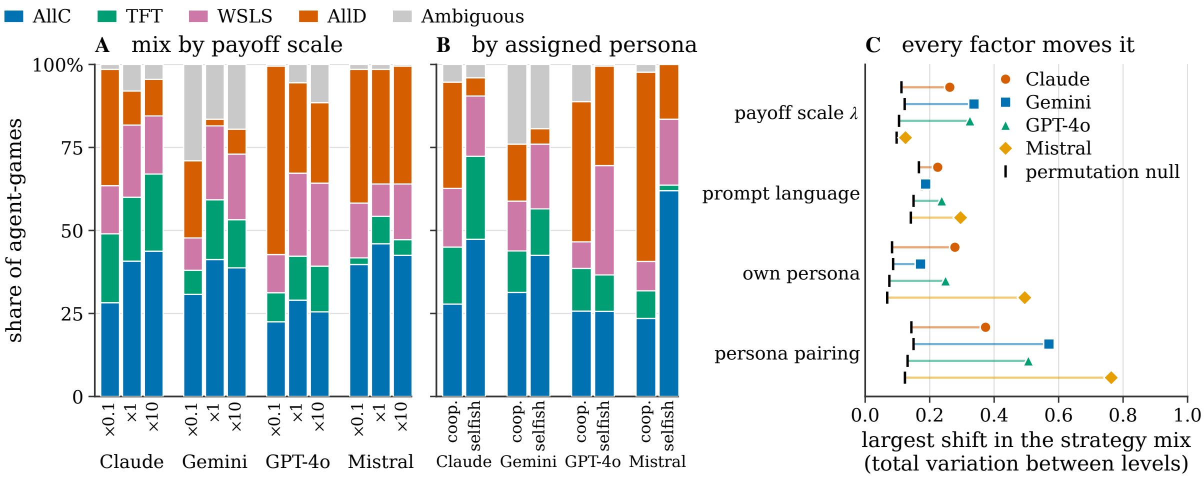

Figure 4: The distribution over strategies moves with changes that leave the game unchanged. (**a**) The strategy mix at each payoff scale, per model. Multiplying every cell by a positive constant is a positive affine rescaling of a von Neumann–Morgenstern utility, so the three games are one game and the predicted shift is exactly zero. The AllD share of GPT-4o falls from 0.57 to 0.24 across it. (**b**) The same mix by assigned persona. Mistral Large told it is cooperative plays AllD in 57% of its games and AllC in 24%; told it is selfish, the same model gives 17% and 62%. The instruction moves the mix against its own meaning. (**c**) Whether the mix moves at all, for every model and factor. Markers give the largest total-variation distance between the mixes at two levels, and ticks give the 95th percentile of a permutation null that shuffles the factor label across whole dyads. All sixteen comparisons exceed their null.

The persona manipulation produces the largest shifts in the study, and it is the most interesting because it runs backwards. Mistral Large told it is cooperative carries the AllD label in 57.0% of its games and AllC in 23.5%; told it is selfish, the same model carries AllD in 16.5% and AllC in 62.0%. Claude moves from 32.0% AllD and 27.8% AllC to 5.5% and 47.3%. In every model the selfish persona produces less unconditional defection than the cooperative one, and in three of the four it produces more unconditional cooperation. GPT-4o is the exception only in that its AllC share stays flat at 0.26 while its AllD share falls from 0.42 to 0.30. Resolving the manipulation into the four ordered pairings moves the mix by 0.373 to 0.763, against own-persona shifts of 0.172 to 0.495. An agent is never told its opponent's persona, so the excess of the pairing effect over the own-persona effect is a response to the interaction rather than to the instruction.

Both directions fit a single reading, which we offer as the simplest account rather than as a demonstration. The models may treat the displayed numbers as rewards to be maximised rather than as penalties to be minimised. Under that reading they match the matrix to the canonical reward-framed dilemma, in which the largest entry is the prize. Option A then yields the symmetric pair $(6, 6)$, and Option B can yield the single largest number, 10. An agent reaching for a mutually good outcome therefore takes A, and an agent reaching for the largest attainable number takes B, which is what the persona columns show. Raising λ makes that maximum easier to notice, which is the direction the scale panel shows. Separating the reward reading from an option-ordering account requires restating the identical matrix in explicit reward form, an experiment we did not run.

Whichever account is right, the consequence for attribution is the same and does not depend on it. A strategy census is not a measurement of the agent unless it stays fixed under

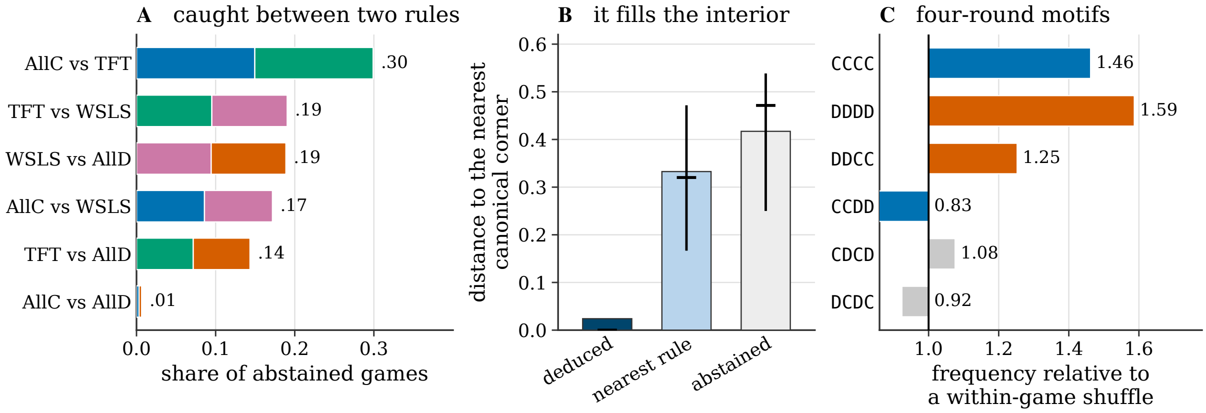

Figure 5: The abstention set is structured, and the structure lies outside the vocabulary. (**a**) Which two rules the posterior is split between. Each bar is halved between the pair. The network is not spreading its confidence over four labels; it is caught between two, and TFT is in 63% of the ties. (**b**) Distance to the nearest canonical corner of the reactive square $p = P(C \mid \text{opponent } C)$, $q = P(C \mid \text{opponent } D)$, in which AllC, TFT and AllD are corners. Bars are means and whiskers the interquartile range. Deduced play sits on a corner by construction; abstained play sits in the interior. (**c**) Frequency of four-round windows of the focal player's own actions, relative to a shuffle of that player's own actions within its own game. Runs and a single change of regime are enriched; alternating play, the pattern a memory-one rule cannot express, is not.

transformations that leave the game alone, and on this corpus it does not.

3.5 The anatomy of what the read-out will not name

A quarter of the corpus leaves the read-out with no label. The easy dismissal is that the rest is noise, and answering that charge decides whether the vocabulary is too small or the wrong shape.

The abstention set is not a shrug (figure 5a). The posterior in the abstained games is not spread over four labels but split between two. The top pair is AllC versus TFT in 29.9% of cases, TFT versus WSLS in 19.0%, WSLS versus AllD in 18.9%, AllC versus WSLS in 17.2% and TFT versus AllD in 14.3%. AllC versus AllD, the only pair with no rule between them, takes 0.6%. TFT appears in 63.2% of the ties. The network is refusing to choose between neighbouring rules, not failing to see anything at all.

The play is not near the vocabulary either (figure 5b). In the reactive square, where AllC, TFT and AllD are corners, the mean distance to the nearest corner is 0.024 for deduced games, 0.333 for games given a nearest rule and 0.417 for abstained games. The medians are 0.000, 0.320 and 0.471, on a square whose corners are at most 1.41 apart. The ordering is the point. A nearest-neighbour label goes to play a third of the way to the nearest canonical point, and the games the read-out will not name are further still.

The motif analysis says what kind of structure is there (figure 5c), and it contradicts the most natural guess. A plausible hypothesis for unnameable play in a two-action game is alternation, since a strictly alternating sequence is invisible to a memory-one rule. Alternation is not what we find. Relative to a within-game shuffle that preserves each player's own cooperation rate, the four-round windows enriched in the abstention set are the runs and the single switch: DDDD at a lift of 1.59, CCCC at 1.46 and DDCC at 1.25. Alternation stays at chance, with CDCD at

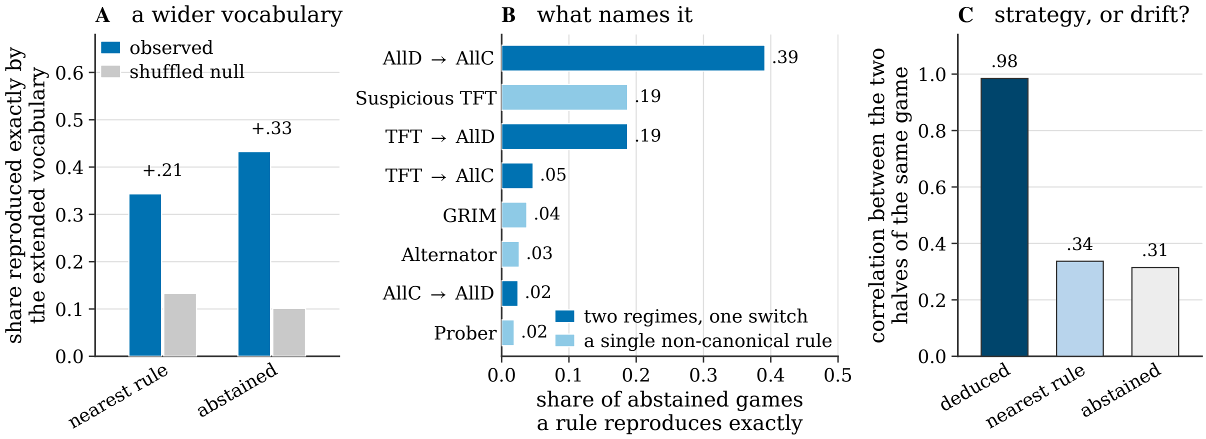

Figure 6: What the abstained play is, and whether it is a strategy. (**a**) Share of games reproduced exactly by the extended vocabulary of fifteen named non-canonical rules plus 54 two-phase clock rules, against the permutation null that shuffles the focal player's own actions. The wider vocabulary earns 0.33 of coverage above chance on the abstained games and 0.21 on the games given a nearest rule. (**b**) The rules that reproduce an abstained game exactly. Two thirds of the fits are two-phase, the median switch falls at round two, and most switches move towards cooperation. (**c**) Correlation between the cooperation rate of the first and second halves of the same game. Play that a rule reproduces is coherent by construction; the abstention set is barely less coherent than play the classifier labelled confidently, and clustering it finds no separated modes.

1.08 and DCDC at 0.92. Only 9.4% of abstained games alternate more than their own base rate implies by a margin of 0.15 or more, and pooled over the set the alternation excess is negative, −0.036 ([−0.046, −0.027]): these agents keep repeating themselves. Whatever the abstention set is made of, it is made of long runs with a change of regime inside them.

3.6 Two-phase play, opening with a probe

Widening the vocabulary answers the question the canonical four cannot (figure 6a). The 69 candidates of §2.4 reproduce 43.3% of the abstained games exactly, against a shuffled null of 10.1%, an excess of 33.2 percentage points. On the games the classifier was confident about, the same vocabulary reproduces 34.4% against a null of 13.3%. Per model the abstained coverage is 27.3% for Claude, 55.7% for Gemini, 39.3% for GPT-4o and 59.1% for Mistral, against nulls of 9.6%, 14.0%, 6.7% and 9.7% in the same order. For 90.2% of the abstained games the nearest rule in the wide library is not one of the canonical four.

What lands is specific (figure 6b). Of the 534 abstained games that the extended vocabulary reproduces exactly, 66.9% are two-phase rules with a single switch, and the median switch falls at round two. The single most frequent fit is AllD followed by AllC, at 39.1% of the exact fits. Next comes suspicious TFT, which opens with defection and reciprocates afterwards, at 18.7%. Then follow TFT then AllD at 18.7%, TFT then AllC at 4.7%, grim trigger at 3.7%, the alternator at 2.6%, AllC then AllD at 2.4% and prober at 1.9%.

Two features of that list deserve separating from the arithmetic. The first is its direction. Of the fitted switches, 65.5% end in a more cooperative regime than they began, against 31.7% that end in a less cooperative one and 2.8% that end in TFT. Backward induction on a finite horizon predicts the opposite, and the switches do not fall where an endgame effect would put them, but at a median of round two. Whatever produces them is an opening phenomenon.

The second feature is that three of the four most frequent fits share a shape: an opening defection that is not sustained. AllD-then-AllC, suspicious TFT and prober all begin by defecting and then hand control to something else, and together they account for 59.7% of the exact fits. Read together with the persona inversion of §3.4, the natural description is a probe. The agent uses the opening move to sample the opponent rather than to commit to a line of play, and the strategy proper starts on round two or three. That shape is not in the canonical vocabulary at any width, because it is not memory-one; it is a rule about the round index. It is also not what any of the usual archetypes would predict, and it is our main answer to what LLM play in this corpus consists of when it is not constant.

The mechanism behind the failure of the memory-one class is directly visible. Scoring nested predictors of the next action on identical rows, all four models select memory-two by BIC. The improvement over a memoryless baseline goes from -1685 to -2399 for Claude, -4334 to -4792 for Gemini, -3684 to -4865 for GPT-4o and -6652 to -9158 for Mistral. Allowing the memory-one rule to change with the phase of the game instead, an extension of the same size in the direction of time rather than of history, is worse than memory-two for every model. The rows are pooled within a model, and a mixed population of memory-one agents can look memory-two once pooled, so this test is evidence that the corpus is not memory-one rather than that any single agent-game is not.

One limit applies to the wider vocabulary itself and bounds how much of the above can be read as identification. Over ten rounds a run of play rarely visits all four conditioning states, so rules that differ only in an unvisited state cannot be told apart. Of the 4,800 agent-games, 2,608 match none of the 32 deterministic memory-one rules, and of those that do match, most are matched by four or eight rules at once. Exactly 89 games, or 1.9% of the corpus, are pinned to a single rule. A coverage figure for the wider class is therefore a statement about a *set* of rules and not about a rule, and the ten-round horizon is what makes it so.

3.7 Strategy, or drift

Whether the unnameable play is a strategy the vocabulary is missing, or no strategy at all, has an answer the data can bound, and the answer is both (figure 6c).

Play generated by a fixed rule is coherent across the game by construction, and the deduced games confirm that the instrument works: there the correlation between the cooperation rates of the two halves of a game is 0.985. The same correlation is 0.337 among games given a nearest rule and 0.315 among abstained games. The first half of an abstained game barely predicts its second half, and, importantly, neither does the first half of a game the classifier was confident about. Low coherence is therefore not a property of the abstention set alone. It is a property of all LLM play in this corpus that no rule reproduces.

Clustering finds no fifth archetype. In memory-one profile space the silhouette of a k-means partition of the abstention set falls from 0.378 at $k = 2$ to 0.319 at $k = 8$, and it never rises above its value at $k = 2$. The data therefore prefer no number of clusters. The bootstrap adjusted Rand index is high at small k, 0.93 at $k = 2$ and 0.94 at $k = 3$. That index, however, measures whether a partition is reproducible rather than whether it is separated, and a single long cloud cut in half reproduces perfectly. The three-cluster solution is interpretable, with one mode that persists and imitates (self-persistence 0.80, opponent-matching 0.82, 56% of the set), one near 0.5 on every coordinate (31%) and one that is anti-imitative (opponent-matching 0.12, 13%).

These three modes are regions of a continuum rather than separated clusters.

The verdict is therefore split, and we state it as such. About 43% of the abstention set is a rule outside the canonical four, most often a two-phase rule with an opening probe, and that share is more than four times its shuffled null. The rest of that set, and the majority of the play the classifier labels confidently, is within-game drift that no fixed rule will name, because there is no fixed rule there to find.

4 Discussion

4.1 A label needs to say where it came from

The central result is a gap between two quantities that are usually reported as one. Reactive strategy labels go to 30.3% of the agent-games in this corpus, and reactive rules are deductively established in 2.0% of them. Both numbers are correct. They answer different questions, and only the second supports the conclusions that a strategy name is normally used to license.

The distinction is not a matter of wording, because the two shares behave differently under everything else in the paper. The deduced share does not depend on where the confidence floor is placed, while the inferred share does. The deduced share is dominated by constant play, so it barely changes when we permute a player's own actions, which is why a null of that kind tests the reactive part of the vocabulary and nothing else. And the deduced share is the only one to which the validation of §3.1 transfers, because it is the only one for which "correct" is defined without reference to a training distribution.

The remedy is cheap. An exact matcher costs nothing next to a classifier, and reporting its output as a separate column costs nothing at all. We propose that any strategy attribution for an artificial agent carry three shares, deduced, several-rules-fit and nearest-neighbour, together with the mean number of rounds on which play departs from the rule it was assigned. On this corpus those numbers are 25.5%, 8.3%, 66.2% and 1.97 to 2.49 rounds out of ten.

4.2 The vocabulary is the wrong shape, not the wrong size

The natural repair for a vocabulary that does not fit is to enlarge it, and the way that repair fails here is informative. Going from four rules to all 32 deterministic memory-one rules, and then to two of them in sequence, still leaves a large part of the corpus unfitted. The wide library that does earn 33 points over its null earns them mostly with rules that are not memory-one at all but two-phase. Such a rule keeps one regime for the opening and another for the rest, with a median switch at round two. The BIC comparison points the same way, selecting memory-two over memory-one for every model, and the motif analysis rules out the obvious alternative, since alternation sits at chance while runs and single switches are enriched.

Together these results say that the class is the wrong shape rather than too small. A memory-one description asks what an agent does after each joint outcome. These agents are doing something that depends on where in the game they are as well as on what just happened, and the dependence is concentrated in the first two or three rounds. A vocabulary for LLM agents should therefore be built from phases and probes rather than by adding more rules to a memory-one enumeration.

The direction of the switches sharpens the point. Two thirds of them run towards cooperation, and they cluster at the opening rather than at the endgame. The standard prediction for a finite horizon is a slide towards mutual defection; what this corpus contains is an opening defection that the agent abandons. That pattern is closer to exploration than to backward induction, and a population model that assigned these agents a fixed strategy would get both the sign and the timing wrong.

4.3 Consequences for evolutionary models of hybrid populations

Work that models populations in which artificial and human agents interact [2, 30, 31] usually assigns agents strategies from a known family and computes the resulting dynamics. Three of our results bear on that programme. First, the assignment is mostly not available: a replicator equation needs each agent placed at a point, and here that point is deductively determined for a quarter of agent-games and is a constant rule in 92% of those. Second, the assignment is not stable under transformations the dynamics cannot see. Rescaling all payoffs leaves the replicator equation identical and moves the mix by up to 0.34, so two population models that a game theorist would call the same model predict differently about the same agent. Third, the individual game is not coherent enough to carry a label at all, with a between-half correlation of 0.32 outside the deduced set. What these three results support is a distribution over strategies whose support changes with framing, not a strategy.

4.4 Relation to the strategy-attribution literature

The closest neighbours are studies that recover archetypes from LLM play: Fontana et al. [17] over long horizons, Akata et al. [15] as stable model-specific signatures, and Lorè and Heydari [19], who separate game structure from contextual framing and find that framing carries comparable weight. Our results are not in tension with those measurements, and we do not claim their labels are wrong. What we add is the denominator, that is, how often such a label is actually proved. An archetype recovered from LLM play is a nearest-neighbour statement unless a deductive stage says otherwise, and on our corpus it says otherwise for 2.0% of reactive labels. The framing sensitivity that Lorè and Heydari [19] document for outcomes, we document one level up, for the strategy distribution itself, under a change whose null is a theorem rather than a default. Our results also connect the machine-behaviour programme [3] to the framing literature in human decision making [32, 33], with one instructive difference. Human framing effects map to a compact set of psychological mechanisms, whereas what moves here is the identity of the strategy that a standard read-out reports, and it moves under a change of units.

4.5 Limitations

Four limitations bound these conclusions, and the first two would change the numbers rather than the interpretation.

The horizon is ten rounds, which bounds §3.6 in particular. It under-identifies the memory-one vocabulary, so most games matched by the wider class are matched by a set of rules rather than by one, and only 1.9% of the corpus is pinned to a single one of the 32. It also gives a two-segment fit enough freedom that a shuffled null already reaches 0.10, which is why we report every coverage figure against that null. Finally, it compresses any endgame effect, so the absence of endgame switching is weaker evidence than the presence of opening switching. The horizon does not, however, bound §3.3, and we would put the two controls that establish this to a doubtful reader first. At ten rounds the same matcher pins a unique answer on 80.8% of TFT-generated and 79.6% of WSLS-generated synthetic play, as often as on the constant rules. And the deduced-reactive share barely moves when we restrict the corpus to the two thirds of agent-games whose opponent history makes the distinction visible.

The model set is four, a small sample of a fast-moving population, and nothing here establishes that the pattern generalises to models with explicit reasoning procedures. Testing the read-out on reasoning models is the single most valuable extension, and we regard these results as establishing the instrument rather than settling the question.

Two confounds with model identity remain. The Gemini arm knew the round count and the others did not, so that column mixes model identity with horizon knowledge. Sampling temperature is likewise fixed by the provider default, so no between-model ordering of a coverage statistic is interpretable and we claim none. Neither confound is constant across the levels of any manipulation, so neither can generate the within-model effects of §3.4.

Finally, the confidence floor is a convention at this horizon rather than a valley in the distribution. The reward-reading account of the persona and scale effects (§3.4) is likewise an inference from converging evidence rather than a controlled demonstration. For the first limitation we report the risk-coverage curve and the census at three floors. For the second we identify the relabelling experiment that would settle it.

4.6 Conclusion

Four language models played 2,400 iterated prisoner's dilemmas. A read-out that separates deduction from inference, and that is allowed to abstain, establishes a canonical strategy for a quarter of the resulting agent-games, and 92% of that quarter is unconditional cooperation or unconditional defection. Tit-for-tat and win-stay lose-shift, the rules that carry the theoretical interest of the game, are named fifteen times more often than they are demonstrated, and where they are named, play departs from them on more than two rounds in ten. The distribution over strategies is in any case not a property of the agent: it moves beyond a permutation null under every change we apply, including a rescaling of the payoffs that cannot change the game. Mining the quarter that the read-out will not name recovers a structure the canonical vocabulary has no room for: a two-phase rule that switches once at a median of round two and runs towards cooperation. Behind that structure lies a majority of play that is drift rather than any rule at all.

The practical recommendation follows from the first result alone, and it costs nothing to adopt. Run the deductive stage before the classifier, let the read-out say "no canonical rule explains this", and publish the three shares and the departure count alongside any strategy label. Until that practice is routine, the field risks building an evolutionary game theory of artificial agents on names that were assigned rather than earned.

Data accessibility

The full corpus of 2,400 dyads, the analysis pipeline that produces every figure and table, the trained read-out checkpoint and the supplementary figure suite are available at [REPOSITORY URL]. Every number appearing in a figure is also written to a machine-readable CSV file in the same repository. We will deposit a permanent archive with a DOI on acceptance.

Competing interests

The authors declare no competing interests.

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